

Exact Carry Geometry for Abelian-Square Constraints under Partial Uniform Block Assignment

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Abstract

We study Abelian-square constraints in a constant-length block coding when all source-letter images but one have already been assigned. If the common block length is L and a candidate Abelian square has half-period K , write $K = qL + r$, $0 \leq r < L$. The three equally spaced cutpoints induce two binary carries. Keeping both the carry pair and the distinction $q = 0$ versus $q \geq 1$ yields six exact physical cutpoint domains.

Projecting the Parikh second-difference equation onto the unresolved block role, and enforcing equality of occurrence bits when two cutpoints lie in the same macro block, gives exactly 34 physically realizable domain/mask patterns. Quotienting these patterns by equality of their complete reduced support sets gives exactly 19 stable support families for every $L \geq 5$. We derive closed cardinality formulas for all nineteen families and prove pairwise distinctness symbolically, without a finite-range assumption.

The assigned blocks enter only through affine target values. Consequently, for a prescribed Parikh profile of the unresolved block, every reduced support signature has an exact finite reachable-target set; nonmembership gives an exact feasibility obstruction for a single target-loaded window. A length-40 case study illustrates how the support classification interfaces with finite subset gates and fixed-profile prefix reachability. The structural classification is independent of that case study and provides a finite support compiler for partially assigned uniform Abelian-square constraints.

1. Introduction

Two words are Abelian equivalent when they have the same Parikh vector. An **Abelian square** is a factor UV satisfying

$$|U| = |V|, \quad \Psi(U) = \Psi(V).$$

For a concrete finite word, one may certify Abelian-square avoidance by checking candidate factors directly. In a uniform block coding, however, a useful intermediate structure appears before the coding is fully assigned.

Let a source word over a finite macro alphabet be coded by blocks of common length L . Assume that all block roles except one have fixed images. For a candidate square, the three cutpoints at the beginning, midpoint, and end of the factor contribute a Parikh second difference. After the already assigned block contributions are moved to the target side, only

prefix Parikh vectors of the unresolved role remain on the support side. The problem studied here is:

Which reduced unresolved-support equations can physically occur?

A typical support has the form

$$x_i - 2x_j + x_k,$$

with one or more terms absent when the corresponding cutpoint lies in an assigned block. The essential point is that the three local depths (i, j, k) are not arbitrary: they are linked by Euclidean carry geometry, and the occurrence mask is constrained whenever two cutpoints lie in the same macro block.

1.1 Main result

The classification proved in this paper can be summarized as

three cutpoints $\longrightarrow$ 6 physical domains $\longrightarrow$ 34 realizable domain/mask patterns $\longrightarrow$ 19 complete support families

Theorem A.

Let $L \geq 5$, and let $H : \Gamma \rightarrow \Sigma^L$ be an L -uniform coding in which exactly one source role $X \in \Gamma$ is unresolved. For every candidate Abelian square of half-period $K \geq 2$:

1. its three local cutpoints belong to exactly one of six physical carry domains

$$Z_s, P_t, M_t, Z, P, M;$$

2. across all physically consistent occurrence masks there are exactly 34 realizable domain/mask patterns;
3. identifying two such patterns precisely when their **complete reduced X -support sets** are equal yields exactly 19 equivalence classes.

The nineteen classes are stable for every $L \geq 5$, have explicit cardinalities, and are pairwise distinct for all $L \geq 5$.

The theorem is proved as Theorem 5.1. It concerns the **support geometry** of partially assigned constraints. It does not assert that checking nineteen objects certifies all periods, and the nineteen families are not automaton states.

1.2 Support and target

The classification separates a target-loaded window into

$$\boxed{\text{support signature}} \quad + \quad \boxed{\text{affine target from assigned data}}.$$

This separation is exact. Once a total Parikh profile ρ is prescribed for the unresolved block, a fixed support signature σ has a finite reachable set $\mathcal{R}_\sigma(\rho)$. For one declared window, membership of its target in this set is exactly the existence question for an ordering of that profile. This consequence is stated in Section 7.

1.3 Contributions and organization

The paper contributes:

1. an exact six-domain description of three equally spaced cutpoints in a uniform block system;
2. the physically consistent reduction to 34 domain/mask patterns;
3. the exact quotient of those patterns into 19 complete reduced support families;
4. closed cardinality formulas and a symbolic all- $L \geq 5$ distinctness proof;
5. an exact profile-level feasibility interface for a target-loaded support signature;
6. finite subset gates and fixed-profile reachability tools illustrating how the support compiler can be used in staged constructions.

Sections 2–5 prove Theorem A. Section 6 positions the result relative to earlier Abelian-power machinery. Section 7 gives the profile-level feasibility corollary. Sections 8–10 give a length-40 case study and certificate mechanisms. Section 11 records scope, reproducibility, and conclusions. Detailed counting formulas and the secondary finite comparison are placed in the appendices.

2. Preliminaries

Let Γ be a finite **source (macro) alphabet** and Σ a finite **output alphabet**. Let

$$H : \Gamma \rightarrow \Sigma^L$$

be an L -uniform coding. We allow H to be only **partially assigned**: all source letters except one role $X \in \Gamma$ have fixed images, while the word $H(X)$ remains unresolved.

For a word w , write $\Psi(w)$ for its Parikh vector. For the unresolved block define formal prefix states

$$x_i = \Psi(H(X)[0..i]), \quad 0 \leq i \leq L,$$

with

$$x_0 = 0.$$

Consider a candidate abelian square beginning at position s with half-period K . Its cutpoints are

$$t_0 = s, \quad t_1 = s + K, \quad t_2 = s + 2K.$$

Write the unique uniform-block decomposition

$$t_j = b_j L + i_j, \quad 0 \leq i_j < L.$$

Let

$$\chi(b) = \begin{cases} 1, & \text{if macro block } b \text{ has role } X, \\ 0, & \text{otherwise.} \end{cases}$$

The abelian-square condition is the prefix second difference

$$P(t_0) - 2P(t_1) + P(t_2) = 0,$$

where $P(t)$ is the Parikh vector of the coded prefix of length t . Moving every assigned-block contribution to the target side leaves the unresolved support

$$\sigma = \text{red}(\chi(b_0)x_{i_0} - 2\chi(b_1)x_{i_1} + \chi(b_2)x_{i_2}). \quad (2.1)$$

Here **red** removes x_0 , combines equal depths, and deletes zero coefficients. The zero signature is legitimate: it means that the candidate window is decided entirely by assigned data.

A **support family** will mean the complete set of reduced signatures obtained from one geometric domain and one physically consistent occurrence mask.

3. Euclidean carry geometry

Write

$$K = qL + r, \quad 0 \leq r < L. \quad (3.1)$$

Define the two carry bits

$$c_j = \left\lfloor \frac{i_j + r}{L} \right\rfloor \in \{0, 1\}, \quad j = 0, 1. \quad (3.2)$$

Then

$$i_{j+1} = i_j + r - Lc_j \quad (3.3)$$

and

$$b_{j+1} - b_j = q + c_j. \quad (3.4)$$

Thus q counts the complete block lengths in the half-period, while each c_j records whether the remainder r crosses one additional block boundary at the j -th step. This separation is the reason the short $q = 0$ cases must be kept distinct from their full-domain analogues.

Put

$$g_1 = b_1 - b_0, \quad g_2 = b_2 - b_1.$$

The macro curvature is therefore

$$\kappa = g_2 - g_1 = c_1 - c_0 \in \{-1, 0, +1\}. \quad (3.5)$$

The absolute arithmetic-progression identity

$$t_0 - 2t_1 + t_2 = 0$$

also gives

$$i_0 - 2i_1 + i_2 = -\kappa L. \quad (3.6)$$

Setting

$$a = i_0, \quad \eta = i_1 - i_0$$

gives the local normal form

$$i_0 = a, \quad i_1 = a + \eta, \quad i_2 = a + 2\eta - \kappa L, \quad (3.7)$$

together with

$$K = g_1 L + \eta. \quad (3.8)$$

No approximation is involved in these identities.

Lemma 3.1 — six physical domains

For $L \geq 5$ and $K \geq 2$, every candidate square belongs to exactly one of

$$Z_s, \quad P_t, \quad M_t, \quad Z, \quad P, \quad M.$$

The full local triple sets are

$$Z = \{(u, v, w) : u + w = 2v, \ 0 \leq u, v, w < L\}, \quad (3.9)$$

$$P = \{(u, v, w) : u + w = 2v - L, \ 0 \leq u, v, w < L\}, \quad (3.10)$$

$$M = \{(u, v, w) : u + w = 2v + L, \ 0 \leq u, v, w < L\}. \quad (3.11)$$

The short same-block zero-curvature domain is

$$Z_s = \{(a, a + \eta, a + 2\eta) : \eta \geq 2, \ a + 2\eta < L\}. \quad (3.12)$$

The domains P_t and M_t are respectively the $q = 0$ positive- and negative-curvature cases.

Proof The pair $(q, c_0 c_1)$ exhausts all possibilities. When $q = 0$ and $c_0 c_1 = 00$, all cutpoints lie in the same block and $K = \eta \geq 2$, giving Z_s . The cases $q = 0, 01$ and $q = 0, 10$ give the truncated positive and negative domains. Equal carries outside the same-block case give Z . Unequal carries give P or M according to the sign of $c_1 - c_0$. Equations (3.9)–(3.12) follow from (3.7). $\square$

domain	quotient / carries	macro-block relation	curvature
Z_s	$q = 0, c_0 c_1 = 00$	$b_0 = b_1 = b_2$	$\kappa = 0$
P_t	$q = 0, c_0 c_1 = 01$	$b_0 = b_1 < b_2$	$\kappa = +1$
M_t	$q = 0, c_0 c_1 = 10$	$b_0 < b_1 = b_2$	$\kappa = -1$
Z	$q = 0, 11 \text{ or } q \geq 1, c_0 = c_1$	$b_0 < b_1 < b_2$	$\kappa = 0$
P	$q \geq 1, c_0 c_1 = 01$	$b_0 < b_1 < b_2$	$\kappa = +1$
M	$q \geq 1, c_0 c_1 = 10$	$b_0 < b_1 < b_2$	$\kappa = -1$

The six domains distinguish local carry data from the quotient regime $q = 0$ versus $q \geq 1$.

Figure 1: The six physical domains. The carry pair fixes the local curvature, while the quotient regime distinguishes same/truncated cases from full ones.

Worked example

Take $L = 5$, $K = 2$, and let the first cut have local depth $i_0 = 0$. Then $q = 0$, $r = 2$, and the recurrence gives

$$(i_0, i_1, i_2) = (0, 2, 4), \quad (c_0, c_1) = (0, 0).$$

All three cutpoints lie in one macro block, so this is a Z_s window. If that block has the unresolved role, the raw support is

$$x_0 - 2x_2 + x_4,$$

which reduces to $-2x_2 + x_4$ because $x_0 = 0$. This small example is the simplest nonempty member of the stable same-block family.

Lemma 3.2 — one-point truncation

The truncated nonzero-curvature domains differ from the corresponding full domains by exactly one local triple:

$$P_t = P \setminus \{(L-2, L-1, 0)\}, \quad (3.13)$$

$$M_t = M \setminus \{(L-1, 0, 1)\}. \quad (3.14)$$

Proof For P , write $i_1 - i_0 = \eta$. The $q = 0$ condition gives $K = \eta \geq 2$, whereas the full domain permits $\eta = 1$. Substituting $\eta = 1$ into the local inequalities forces $a = L-2$, giving $(L-2, L-1, 0)$.

For M , the smallest possible η is $1-L$, realized only by $(L-1, 0, 1)$. In the $q = 0$ truncated realization the period condition is $L + \eta \geq 2$, which removes precisely that point. $\square$

4. Physical role projection

For a candidate window, write

$$\chi_j = \chi(b_j) \in \{0, 1\}, \quad j = 0, 1, 2,$$

where $\chi_j = 1$ means that the macro block containing the j -th cutpoint has the unresolved role X . The three bits are not always independent: when two cutpoints lie in the same macro block, their bits must agree.

For a mask $\chi = (\chi_0, \chi_1, \chi_2)$, define

$$\pi_\chi(u, v, w) = \text{red}(\chi_0 x_u - 2\chi_1 x_v + \chi_2 x_w), \quad (4.1)$$

where **red** removes x_0 , combines equal depths, and deletes zero coefficients. For a physical domain D , the corresponding **complete support set** is $\pi_\chi(D)$.

Lemma 4.1 — exactly 34 physically realizable domain/mask patterns

For $L \geq 5$:

- Z_s admits 000, 111;
- P_t admits 000, 001, 110, 111;
- M_t admits 000, 100, 011, 111;
- each of Z, P, M admits all eight binary masks.

Hence

$$2 + 4 + 4 + 8 + 8 + 8 = 34. \quad (4.2)$$

Every one of these 34 patterns has a geometric realization for $L \geq 5$.

Proof In Z_s , all three cutpoints lie in the same macro block, so all three occurrence bits agree. In P_t , the first two cutpoints share a block, hence $\chi_0 = \chi_1$; in M_t , the last two share a block, hence $\chi_1 = \chi_2$. In the full domains Z, P, M , the three cutpoints can lie in three distinct macro blocks, so every mask is realizable. Explicit interior points exist for each domain once $L \geq 5$. $\square$

The number 34 counts physical **domain/mask cases**. The number 19 below counts equality classes of **complete support sets**. These are different objects.

Lemma 4.2 — truncation and support rescue

The truncated nonzero-curvature domains are

$$P_t = P \setminus \{p^+\}, \quad p^+ = (L-2, L-1, 0), \quad (4.3)$$

and

$$M_t = M \setminus \{p^-\}, \quad p^- = (L-1, 0, 1). \quad (4.4)$$

Under the physically realizable masks, the deleted point changes the complete support set in exactly one case on each side:

$$\pi_{110}(P_t) = \pi_{110}(P) \setminus \{x_{L-2} - 2x_{L-1}\}, \quad (4.5)$$

$$\pi_{011}(M_t) = \pi_{011}(M) \setminus \{x_1\}. \quad (4.6)$$

All other nonempty truncated support images agree with their full-domain counterparts.

Proof For the positive side, $P_t = P \setminus \{p^+\}$.

- Under mask 001,

$$\pi_{001}(p^+) = x_0 = 0,$$

and 0 is already realized inside P_t , for example by $(L-4, L-2, 0)$.

- Under mask 111,

$$\pi_{111}(p^+) = x_{L-2} - 2x_{L-1}.$$

The point

$$(0, L-1, L-2) \in P_t$$

gives the same reduced signature.

- Under mask 110, the same deleted point gives $x_{L-2} - 2x_{L-1}$. In a 110 projection, the coefficient -2 fixes $v = L-1$ and the coefficient $+1$ fixes $u = L-2$; hence any realization is p^+ itself. No rescue is possible.

For the negative side, $M_t = M \setminus \{p^-\}$.

- Under mask 100,

$$\pi_{100}(p^-) = x_{L-1},$$

which is already realized by $(L-1, 1, 3) \in M_t$.

- Under mask 111,

$$\pi_{111}(p^-) = x_{L-1} + x_1,$$

and

$$(1, 0, L-1) \in M_t$$

gives the same reduced signature.

- Under mask 011, $\pi_{011}(p^-) = x_1$. Producing this singleton forces $v = 0$ and $w = 1$; the domain equation then forces $u = L-1$, so the deleted point is the unique realization.

Thus each truncated side contributes exactly one genuinely new support set. $\square$

4.1 Outer reversal

For each full domain $D \in \{Z, P, M\}$, outer reversal

$$(u, v, w) \longmapsto (w, v, u) \quad (4.7)$$

preserves D . Therefore

$$\pi_{001}(D) = \pi_{100}(D), \quad \pi_{011}(D) = \pi_{110}(D). \quad (4.8)$$

All 000 cases give the same empty support set

$$E = \{0\}. \quad (4.9)$$

After removing E , each full domain therefore contributes at most five nonempty support types. The same-block domain contributes one additional all-active type, and Lemma 4.2 contributes one additional truncated centre–outer type on each nonzero-curvature side. Hence the quotient has at most

$$1 + 1 + 5 + (5 + 1) + (5 + 1) = 19 \quad (4.10)$$

classes. Section 5 proves that all nineteen are distinct.

5. The nineteen support families

To avoid overloading the domain name M , use the following support-type abbreviations:

$$\begin{aligned} O : x_u, \quad C : -2x_v, \quad CO : x_u - 2x_v, \\ OO : x_u + x_w, \quad A : x_u - 2x_v + x_w, \end{aligned} \quad (5.1)$$

up to outer reversal.

The nineteen stable families and their cardinalities are:

family	description	exact cardinality
E	empty support	1
$Z_s\text{-}A$	same-block zero-curvature, all active	$\lfloor (L-3)^2/4 \rfloor$
$Z\text{-}O$	zero curvature, one outer	L
$Z\text{-}C$	zero curvature, centre	L
$Z\text{-}CO$	zero curvature, centre + one outer	$\lfloor L^2/2 \rfloor$
$Z\text{-}OO$	zero curvature, both outers	$\lfloor (L+1)^2/4 \rfloor$
$Z\text{-}A$	zero curvature, all active	$\lfloor (L-1)^2/4 \rfloor + 1$
$P\text{-}O$	positive curvature, one outer	$L-1$
$P\text{-}C$	positive curvature, centre	$\lfloor L/2 \rfloor$
$P\text{-}CO$	positive curvature, centre + outer	$\lfloor L^2/4 \rfloor$
$P\text{-}OO$	positive curvature, both outers	$\binom{\lfloor L/2 \rfloor + 1}{2}$
$P\text{-}A$	positive curvature, all active	$\binom{\lfloor L/2 \rfloor + 1}{2}$
$P_t\text{-}CO$	truncated positive centre + outer	$\lfloor L^2/4 \rfloor - 1$
$M\text{-}O$	negative curvature, one outer	$L-1$
$M\text{-}C$	negative curvature, centre	$\lfloor L/2 \rfloor$
$M\text{-}CO$	negative curvature, centre + outer	$\lfloor L^2/4 \rfloor$
$M\text{-}OO$	negative curvature, both outers	$\binom{\lfloor L/2 \rfloor + 1}{2}$
$M\text{-}A$	negative curvature, all active	$\binom{\lfloor L/2 \rfloor + 1}{2}$
$M_t\text{-}CO$	truncated negative centre + outer	$\lfloor L^2/4 \rfloor - 1$

For $L = 40$, these cardinalities are

$$1, 342, 40, 40, 800, 420, 381, 39, 20, 400, 210, 210, 399, 39, 20, 400, 210, 210, 399. \quad (5.2)$$

Appendix A derives all formulas directly from the domain inequalities.

5.1 Proof of the main classification

Theorem 5.1 (Theorem A).

For every $L \geq 5$, the 34 physically realizable domain/mask patterns quotient, under equality of complete reduced support sets, to exactly the nineteen families in the table above.

Proof Lemma 3.1 gives the exhaustive six-domain partition. Lemma 4.1 gives exactly 34 physically consistent domain/mask cases. Equation (4.8) identifies outer-reversal pairs in each full domain, all 000 masks give the single family E , and Lemma 4.2 shows that each truncated nonzero-curvature domain adds exactly one support set not already present in its full-domain image. Thus (4.10) gives at most nineteen classes.

It remains to show that the nineteen displayed complete support sets are pairwise distinct. This is done symbolically in Section 5.2. Therefore the upper bound is attained. $\square$

5.2 Symbolic pairwise distinctness

For a reduced formal signature

$$\sigma = \sum_{i=1}^{L-1} \alpha_i x_i,$$

define its **depth moment**

$$\mu(\sigma) = \sum_{i=1}^{L-1} i \alpha_i. \quad (5.3)$$

The one-coordinate families are explicit:

$$Z\text{-O} = \{0, x_1, \dots, x_{L-1}\},$$

$$P\text{-O} = \{0, x_1, \dots, x_{L-2}\}, \quad M\text{-O} = \{x_1, \dots, x_{L-1}\}, \quad (5.4)$$

and

$$Z\text{-C} = \{0, -2x_1, \dots, -2x_{L-1}\},$$

$$P\text{-C} = \{-2x_v : \lceil L/2 \rceil \leq v \leq L-1\},$$

$$M\text{-C} = \{0\} \cup \{-2x_v : 1 \leq v \leq \lfloor (L-2)/2 \rfloor\}. \quad (5.5)$$

Hence the three O-families are distinct, the three C-families are distinct, and no O-family equals a C-family.

For the centre–outer families,

$$\mu(Z\text{-CO}) = \{-(L-1), \dots, 0\}, \quad (5.6)$$

$$\mu(P\text{-CO}) = \{-2L+2, \dots, -L\}, \quad (5.7)$$

$$\mu(M\text{-CO}) = \{1, \dots, L-1\}. \quad (5.8)$$

These three moment regimes are disjoint. The truncated/full pairs are separated by the missing witnesses

$$x_{L-2} - 2x_{L-1} \quad \text{and} \quad x_1 \quad (5.9)$$

from Lemma 4.2. Thus the five centre–outer families are pairwise distinct.

For both-outer families, P-OO and M-OO are separated by moment:

$$0 \leq \mu(\sigma) \leq L-2 \quad (\sigma \in P\text{-OO}),$$

whereas

$$L \leq \mu(\sigma) \leq 2L-2 \quad (\sigma \in M\text{-OO}). \quad (5.10)$$

The zero-curvature family is separated from each of them by cardinality:

$$|Z\text{-OO}| = \left\lfloor \frac{(L+1)^2}{4} \right\rfloor \neq \binom{\lfloor L/2 \rfloor + 1}{2} = |P\text{-OO}| = |M\text{-OO}| \quad (5.11)$$

for every $L \geq 5$. This avoids relying on a moment endpoint that can overlap when L is even.

For the all-active families,

$$\mu(Z\text{-A}) = \{0\}, \quad \mu(P\text{-A}) = \{-L\}, \quad \mu(M\text{-A}) = \{L\}, \quad (5.12)$$

while $Z_s\text{-A}$ also has zero moment. The latter pair is separated by zero membership:

$$0 \in Z\text{-A}, \quad 0 \notin Z_s\text{-A}. \quad (5.13)$$

Moreover $Z_s\text{-A}$ is nonempty for $L \geq 5$, since $(0, 2, 4)$ gives $-2x_2 + x_4$.

Finally, the structural groups cannot coincide across types: O-families contain only positive unary signatures; C-families only negative doubled unary signatures; centre–outer families contain genuine $(+1, -2)$ two-depth signatures; OO-families have no negative coefficients; and nonzero A-families contain the second-difference $(+1, -2, +1)$ structure. Together with (5.4)–(5.13), this gives a uniform separator for every one of the $\binom{19}{2} = 171$ unordered family pairs.

Therefore the nineteen complete support sets are pairwise distinct for every $L \geq 5$. $\square$

Corollary 5.2 — minimality under the declared quotient

For every $L \geq 5$, nineteen is minimal under the specific rule

identify two domain/mask patterns exactly when their complete reduced support sets are equal.

This is not a minimal-automaton or optimal-implementation theorem.

5.3 Small L

The equality-class counts for $L = 2, 3, 4, 5, 6, 7$ are

$$9, 15, 19, 19, 19, 19. \tag{5.14}$$

At $L = 4$, however, Z_s is empty, so the nineteen classes are not the stable family list above. The first same-block $K \geq 2$ configuration occurs at $L = 5$, which is the natural lower boundary of Theorem 5.1.

6. Relation to earlier Abelian-power machinery

The second-difference layer itself is not new. Carpi's morphism criteria use prefix Parikh second differences with binary whole-image correction selectors, and the template method of Currie and Rampersad organizes Abelian-power avoidance by finite boundary corrections. Rational carry sequences arising from Euclidean division are likewise standard mechanical-word objects.

The distinction made here is the following partial-assignment operation:

$$\begin{aligned} &\text{known block data} + \text{one unresolved occurrence mask} \\ &\longrightarrow \text{complete reduced unresolved-support family} \end{aligned}$$

6.1 Exact comparison with Carpi's C3 condition

A modern restatement of Carpi's condition C3 asserts, for suitable proper prefixes, the existence of binary selectors δ_j satisfying a whole-image corrected second-difference equation. Under a uniform length L , applying the coordinate-sum functional to the three-cutpoint specialization gives

$$i_0 - 2i_1 + i_2 = L(\delta_0 - 2\delta_1 + \delta_2).$$

Combining this with (3.6) yields

$$\boxed{\delta_0 - 2\delta_1 + \delta_2 = c_0 - c_1 = -\kappa.} \tag{6.1}$$

Thus the scalar second-difference correction is shared prior-art territory.

However, C3 local data do not determine the quotient

$$q = \lfloor K/L \rfloor.$$

Fixing L, r, i_0 fixes the local depths and carry pair independently of q , while changing q changes the number of complete macro blocks crossed between cutpoints. Hence the same

local C3 instance can occur with $q = 0$ and with $q \geq 1$. In particular, local C3 data cannot recover the distinction

$$Z_s \text{ versus } Z,$$

nor the analogous truncated/full nonzero-curvature split.

The six admissible C3 selector triples in the arithmetic-progression specialization therefore do not constitute the six physical domains of Theorem 5.1.

Our positioning is deliberately narrow. We do not claim the $(+1, -2, +1)$ algebra, whole-image correction, Euclidean carry arithmetic, mechanical words, template sieving, or generic finite-state reachability. The theorem established here is the explicit **role-projected partial-assignment classification**: the six physical domains, the physically consistent masks, the complete reduced support sets, and their exact $34 \rightarrow 19$ quotient. Historical priority for broader surrounding machinery is left to the cited literature rather than inferred from the form of the present proof.

7. Exact profile-level target feasibility

The classification in Theorem 5.1 is independent of the internal ordering of the unresolved block. Once its total Parikh profile is fixed, however, every reduced support signature has an exact finite set of attainable values.

Fix

$$\rho \in \mathbb{N}^{|\Sigma|}, \quad |\rho|_1 = L,$$

and let

$$\sigma(x) = \sum_{j=1}^m a_j x_{d_j}, \quad 0 < d_1 < \dots < d_m < L. \quad (7.1)$$

Define

$$\mathcal{R}_\sigma(\rho) = \left\{ \sum_{j=1}^m a_j \Psi(w[0..d_j]) : w \in \Sigma^L, \Psi(w) = \rho \right\}. \quad (7.2)$$

Corollary 7.1 — exact feasibility for one target-loaded window.

Suppose a concrete candidate window reduces, after all assigned contributions have been absorbed into the target, to

$$\sigma(x) = \tau. \quad (7.3)$$

Among unresolved block words of profile ρ , the window can be completed to an Abelian square if and only if

$$\tau \in \mathcal{R}_\sigma(\rho). \quad (7.4)$$

In particular,

$$\tau \notin \mathcal{R}_\sigma(\rho) \implies \text{no ordering of profile } \rho \text{ makes this window an Abelian square.} \quad (7.5)$$

Proof For any depths $0 < d_1 < \dots < d_m < L$, integer vectors $y_{d_1}, \dots, y_{d_m}$ are the prefix Parikh vectors of one word of total profile ρ exactly when

$$|y_{d_j}|_1 = d_j$$

for every j , and

$$0 \leq y_{d_1} \leq \dots \leq y_{d_m} \leq \rho$$

componentwise. Necessity is immediate. Conversely, the successive differences

$$y_{d_1}, y_{d_2} - y_{d_1}, \dots, y_{d_m} - y_{d_{m-1}}, \rho - y_{d_m}$$

are nonnegative integer profiles of the consecutive word segments, so choosing any word with each segment profile realizes the entire chain.

Therefore $\mathcal{R}_\sigma(\rho)$ is exactly the set of values taken by σ over all unresolved block words of profile ρ . Equation (7.3) is the Abelian-square Parikh equation for the declared window after the assigned terms have been moved to the target side, giving (7.4) and (7.5). $\square$

This statement concerns one declared window and one unresolved profile. It is not a certificate for the whole coding, it does not certify all periods, and it does not turn the nineteen support families into nineteen global constraints.

8. Finite subset gates and a length-40 case study

The support compiler is most useful inside a staged construction if already assigned roles are certified completely before new roles are introduced.

Let $w \in \Gamma^\omega$ be a fixed source word and let $S \subseteq \Gamma$ be the set of currently assigned roles.

Lemma 8.1 — finite subset-factor gate

Assume that

$$\text{Fact}(w) \cap S^*$$

has bounded word length. Let $\mathcal{C}_S$ be the factor-maximal elements of this finite language under contiguous-factor containment. Then every output factor of $H(w)$ whose **minimal macro support** uses only roles in S , where the minimal macro support is the shortest contiguous source factor whose coded image covers that output factor occurs inside $H(c)$ for some $c \in \mathcal{C}_S$. Conversely every factor of every $H(c)$, $c \in \mathcal{C}_S$, is a genuine factor of $H(w)$.

Therefore absence of all S -supported Abelian squares is equivalent to checking the finitely many cover images $H(c)$.

Proof If an output factor has minimal supporting macro factor $u \in \text{Fact}(w) \cap S^*$, then u is contained in a factor-maximal $c \in \mathcal{C}_S$. Uniform coding preserves aligned factor containment, so the output factor occurs in $H(c)$. The converse is immediate because each c is an actual source factor. $\square$

If $|c| = m$, then every square contained in $H(c)$ has half-period at most

$$\left\lfloor \frac{mL}{2} \right\rfloor. \quad (8.1)$$

This supplies a natural finite completion bound for each assigned-only stage.

8.1 Source and prescribed profiles

We now illustrate the structural theory on a concrete staged coding. The source alphabet is

$$\Gamma = \{a, b, c, d, e, f\}$$

with the 3-uniform morphism

$$h_6 : \quad a \mapsto ace, \quad b \mapsto adf, \quad c \mapsto bdf, \quad d \mapsto bdc, \quad e \mapsto afe, \quad f \mapsto bce. \quad (8.2)$$

The source language is

$$\text{Fact}(h_6^\omega(a)).$$

The application seeks ternary blocks of length 40. The prescribed Parikh profiles are

role	# a	# b	# c
A	15	14	11
B	11	12	17
C	10	14	16
D	12	10	18
E	13	16	11
F	19	11	10.

(8.3)

They arise from a rank-one lift of the incidence matrix of the shorter Rao–Rosenfeld coding.

Proposition 8.2 — kernel preservation under rank-one lifting

Suppose the columns of a matrix B have common nonzero sum L_0 , and let

$$B' = sB + u\mathbf{1}^T, \quad s \neq 0.$$

Here $\mathbf{1}$ denotes the all-ones column vector of the appropriate dimension. If the common column sum of B' is nonzero, then

$$\ker B' = \ker B.$$

Proof If $Bx = 0$, summing coordinates gives $\mathbf{1}^T x = 0$, and hence $B'x = 0$. Conversely, $B'x = 0$ and the nonzero common column sum give $\mathbf{1}^T x = 0$, after which $sBx = 0$. $\square$

For the application, adding $(10, 10, 10)^T$ to every column raises the block length from 10 to 40 while preserving the relevant kernel.

8.2 Factor-maximal subset covers

For the source language (8.1), independently regenerated maximal covers include

$$\mathcal{C}_{\{A,F\}} = \{faf\},$$

$$\mathcal{C}_{\{A,E,F\}} = \{eafea, fafea\},$$

and

$$\mathcal{C}_{\Gamma \setminus \{C\}} = \{eafea, bdfadfbdfafea, ebdfafeadfbdfafea\}.$$

At $L = 40$ their natural global half-period ceilings are respectively

$$60, \quad 100, \quad 340.$$

These finite gates are application-specific; Lemma 8.1 is the general statement.

9. Upstream compatibility as a fixed support skeleton

Fix blocks E, A with profiles

$$\Psi(E) = (13, 16, 11), \quad \Psi(A) = (15, 14, 11).$$

Let

$$p(i) = \Psi(E[0..i]), \quad x(j) = \Psi(A[0..j]).$$

Examining square windows that end while the A -block is being exposed gives three exact support classes. The words **ternary**, **binary**, and **unary** refer here to the number of unresolved prefix vectors of A that remain after the fixed E -contribution is moved to the target side; they do not refer to the size of the output alphabet.

Proposition 9.1 — $E \rightarrow A$ decomposition

For the endpoint $n = 40 + m$, the constraints are

class	range	forbidden equality
ternary	$2k \leq m$	$x(m) - 2x(m - k) + x(m - 2k) = 0,$
binary	$m < 2k \leq 2m$	$x(m) - 2x(m - k) = p(40) - p(40 + m - 2k),$
unary	$k > m$	$x(m) = \lambda_{2k-m} - 2\lambda_{k-m},$

(9.1)

where λ_r is the Parikh vector of the last r letters of E .

For $L = 40$, the raw support counts are exactly

$$\boxed{361 \text{ ternary}, \quad 419 \text{ binary}, \quad 380 \text{ unary.}} \quad (9.2)$$

The support positions are independent of E ; only the affine targets move with E .

Thus compatibility is a monotone fixed-profile prefix-path problem

$$(0, 0, 0) \longrightarrow (15, 14, 11).$$

Lemma 9.2 — depth-one suffix obstruction

For $2 \leq k \leq 20$, define

$$d_k(E) = \Psi(E[41 - 2k..41 - k]) - \Psi(E[41 - k..40]).$$

Let

$$\text{BLOCKED}(E) = \{\alpha \in \Sigma : d_k(E) = e_\alpha \text{ for some } 2 \leq k \leq 20\}.$$

If

$$\text{BLOCKED}(E) = \Sigma,$$

then no compatible A exists.

The criterion is sufficient, not necessary.

10. Downstream fixed-profile reachability and certificates

Fix an A, E pair and leave the order of the F -block unresolved, with

$$\rho = \Psi(F) = (19, 11, 10).$$

For a candidate $F = f_1 \cdots f_{40}$, write

$$X_i = \Psi(f_1 \cdots f_i), \quad X_0 = 0, \quad X_{40} = \rho.$$

Every candidate Abelian-square window becomes an affine forbidden condition

$$\sum_i \alpha_i X_i \in T \quad (10.1)$$

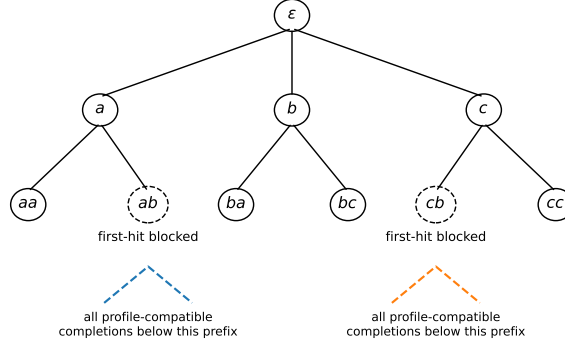
for a finite target set T determined by assigned data.

The profile tree is

$$\mathcal{T}_\rho = \{u \in \Sigma^{\leq 40} : \Psi(u) \leq \rho\}. \quad (10.2)$$

A constraint is **closed** when its largest referenced prefix depth has been reached.

First-hit prefix certificate: schematic



A blocked prefix represents an entire cylinder of complete words; a satisfying word follows a path avoiding all first-hit blocked edges.

Figure 2: Schematic first-hit certificate. A first-hit blocked prefix removes the entire cylinder of profile-compatible complete words extending that prefix; a satisfying word is a root-to-leaf path that avoids every blocked edge.

Theorem 10.1 — path equivalence

A profile-correct F -word satisfies all compiled constraints if and only if its root-to-leaf path in $\mathcal{T}_\rho$ never enters a first-hit blocked edge.

Hence completion is an exact reachability problem, not a property of a particular depth-first implementation.

10.1 First-hit certificate

Let B be the set of bad prefixes whose parents are legal. Then B is prefix-free. If

$$\mathcal{W}_\rho = \{F : \Psi(F) = \rho\}$$

and

$$[u]_\rho = \{F \in \mathcal{W}_\rho : u \preceq F\},$$

then the satisfying set $\mathcal{S}$ obeys the exact partition

$$\mathcal{W}_\rho = \mathcal{S} \dot{\cup} \bigcup_{u \in B} [u]_\rho. \quad (10.3)$$

For $p = \Psi(u)$,

$$|[u]_\rho| = \frac{(40 - |u|)!}{(19 - p_a)!(11 - p_b)!(10 - p_c)!}. \quad (10.4)$$

The total profile space is

$$|\mathcal{W}_\rho| = \frac{40!}{19! 11! 10!} = 46\,305\,405\,961\,214\,400. \quad (10.5)$$

An unsatisfiable instance can therefore be certified by a complete first-hit trie whose blocked cylinders cover (10.5). An independent checker need only verify the profile-admissible transitions and the attached affine killers.

10.2 Frontier sufficiency, and what does not compress

At depth d , let A_d be the historical prefix depths still referenced by an unclosed constraint. These are precisely the old prefix values that a future affine condition may still need. The **frontier state**

$$S_d(u) = (X_d(u), (X_i(u))_{i \in A_d}) \quad (10.6)$$

is sufficient: equal frontier states have equal legal continuation languages.

This is a correctness quotient, not a minimality theorem. In the actual length-40 system, an implementation audit finds

$$A_d = \{1, \dots, d\}$$

for 38 of the 40 depths and

$$\max_d |A_d| = 38.$$

Consecutive prefix differences recover the entire prefix word, so every realized quotient multiplicity is 1. In this concrete system the exact frontier quotient therefore coincides with the legal prefix trie rather than compressing it.

11. Reproducibility, scope, and conclusion

11.1 Reproducibility

The proof of Theorem 5.1 is symbolic. Computational artifacts are retained as a falsification and replay layer rather than as proof premises. Independent checkers enumerate the physical domains and masks, reconstruct the complete support sets, verify the cardinality formulas on finite regression ranges, and compare all family pairs. The symbolic distinctness argument in Section 5.2, not finite enumeration, closes the all- $L \geq 5$ claim.

For the length-40 case study, the reproducibility package records frozen scientific inputs, SHA-256 hashes, deterministic replay commands, independent implementation checks, and literal witnesses for positive completion cases. The project source, proof-replay material, and computational validation artifacts are maintained at <https://github.com/word-structures/combinatorics-on-words-research>.

11.2 Scope

The main simplification is a reduction in **support schemas**, not in the number of candidate square periods. A concrete window is compiled into

$$\text{one support signature} \quad + \quad \text{one affine target}.$$

The resulting limitations are important:

- the nineteen families are not nineteen automaton states;
- Theorem 5.1 does not imply that checking nineteen objects certifies every half-period K ;
- a support family may have many instantiated targets;
- Corollary 7.1 is a single-window, fixed-profile feasibility statement;
- target-loaded simultaneous completion may retain essentially the full prefix history, as Section 10 demonstrates;
- the general Parikh second-difference and whole-image-correction mechanisms belong to prior work; the result here is the role-projected partial-assignment classification.

11.3 Conclusion

For a uniform block system with one unresolved role, the local geometry of an Abelian-square constraint is finite and exact. Euclidean division of the half-period gives six physical cutpoint domains. Repeated-block consistency leaves 34 realizable domain/mask patterns, and equality of complete reduced support sets collapses them to exactly 19 stable families for every $L \geq 5$. Their cardinalities are explicit and their pairwise distinctness is symbolic.

The support/target separation also provides a precise interface to staged construction. For a prescribed unresolved-block profile, a target-loaded window is feasible exactly when its target lies in the reachable set of its compiled support signature. Bounded assigned-only source factors give finite subset gates, and simultaneous target-loaded completion can be represented as fixed-profile prefix reachability.

The broader point is structural: partial assignment need not be treated merely as an incomplete brute-force search. It already carries a reusable combinatorial geometry that can be isolated before concrete targets and search order are introduced.

Appendix A. Cardinality derivations

We sketch all nineteen counts from the exact family descriptions. Throughout, $x_0 = 0$.

A.1 Same-block family

For Z_s -A, each step $\eta \geq 2$ allows

$$0 \leq a \leq L - 1 - 2\eta,$$

hence $L - 2\eta$ starts. Therefore

$$|Z_s\text{-A}| = \sum_{\eta=2}^{\lfloor (L-1)/2 \rfloor} (L - 2\eta) = \left\lfloor \frac{(L-3)^2}{4} \right\rfloor.$$

The reduced signature determines (a, η) , so no further quotient occurs.

A.2 Zero curvature

The diagonal (d, d, d) exists for every $0 \leq d < L$, giving

$$|Z\text{-O}| = |Z\text{-C}| = L.$$

For $Z\text{-CO}$, at fixed centre v ,

$$\max(0, 2v - L + 1) \leq w \leq \min(L - 1, 2v),$$

so there are

$$2 \min(v, L - 1 - v) + 1$$

choices. Summing gives

$$|Z\text{-CO}| = \left\lceil \frac{L^2}{2} \right\rceil.$$

For $Z\text{-OO}$, the unordered outer pair has equal parity. With

$$e = \left\lceil \frac{L}{2} \right\rceil, \quad o = \left\lfloor \frac{L}{2} \right\rfloor,$$

the number of unordered pairs with repetition is

$$\binom{e+1}{2} + \binom{o+1}{2} = \left\lfloor \frac{(L+1)^2}{4} \right\rfloor.$$

For $Z\text{-A}$, every off-diagonal unordered outer pair gives a distinct signature, while all L diagonal triples collapse to the zero signature. Thus

$$|Z\text{-A}| = |Z\text{-OO}| - L + 1 = \left\lfloor \frac{(L-1)^2}{4} \right\rfloor + 1.$$

A.3 Positive curvature

The relation

$$u + w = 2v - L$$

forces

$$\left\lceil \frac{L}{2} \right\rceil \leq v \leq L - 1, \quad 0 \leq u + w \leq L - 2.$$

Every outer depth $0, \dots, L - 2$ occurs, so

$$|P\text{-O}| = L - 1.$$

The possible centre depths give

$$|P\text{-C}| = \left\lfloor \frac{L}{2} \right\rfloor.$$

At fixed v , the mixed family has

$$0 \leq w \leq 2v - L,$$

and the projection is injective, hence

$$|P\text{-CO}| = \sum_{v=\lceil L/2 \rceil}^{L-1} (2v - L + 1) = \left\lfloor \frac{L^2}{4} \right\rfloor.$$

Let $n = \lfloor L/2 \rfloor$. The admissible unordered outer-pair sums have the parity of L and range below L . Counting by sum gives

$$|P\text{-OO}| = \sum_{j=0}^{n-1} (j+1) = \binom{n+1}{2}.$$

The middle depth is uniquely determined by the outer pair, so the same count holds for $P\text{-A}$:

$$|P\text{-A}| = \binom{\lfloor L/2 \rfloor + 1}{2}.$$

Finally, the positive truncation deletes exactly one mixed signature:

$$|P_t\text{-CO}| = \left\lfloor \frac{L^2}{4} \right\rfloor - 1.$$

A.4 Negative curvature

The relation

$$u + w = 2v + L$$

forces

$$0 \leq v \leq \left\lfloor \frac{L-2}{2} \right\rfloor, \quad L \leq u + w \leq 2L - 2.$$

No outer depth is 0, while every $1, \dots, L-1$ occurs, so

$$|M\text{-O}| = L - 1.$$

The centre range, with depth 0 contributing the zero signature, gives

$$|M\text{-C}| = \left\lfloor \frac{L}{2} \right\rfloor.$$

For the centre-outer family,

$$2v + 1 \leq w \leq L - 1,$$

and the projection is injective. Therefore

$$|M\text{-CO}| = \sum_{v=0}^{\lfloor (L-2)/2 \rfloor} (L-1-2v) = \left\lfloor \frac{L^2}{4} \right\rfloor.$$

Reflection

$$(u, w) \mapsto (L-1-u, L-1-w)$$

maps the negative-curvature outer-pair set bijectively to the positive one, hence

$$|M\text{-OO}| = |M\text{-A}| = \binom{\lfloor L/2 \rfloor + 1}{2}.$$

The negative truncation removes exactly the signature x_1 , so

$$|M_t\text{-CO}| = \left\lfloor \frac{L^2}{4} \right\rfloor - 1.$$

Together with $|E| = 1$, these are all nineteen formulas.

Fixed-profile first-hit counting

For a fixed profile $\rho = (\rho_1, \dots, \rho_m)$, write

$$\binom{n}{v} = \frac{n!}{\prod_j v_j!}$$

for the multinomial coefficient when $|v|_1 = n$, where $|v|_1 = \sum_j v_j$. A first-hit bad prefix u with profile $p = \Psi(u)$ eliminates exactly

$$C_\rho(p) = \binom{L - |p|_1}{\rho - p}$$

complete profile words.

If B is the first-hit antichain and $\mathcal{S}$ the satisfying set, then

$$|\mathcal{S}| + \sum_{u \in B} C_\rho(\Psi(u)) = \binom{L}{\rho}. \quad (\text{B.1})$$

This identity is independent of search order.

More generally, if legal prefixes with identical future-relevant frontier data are grouped into a state s of multiplicity $N_d(s)$, then a first-hit blocked transition to child profile p contributes

$$N_d(s)C_\rho(p)$$

to the blocked mass. Summing legal-leaf multiplicity and blocked mass over all layers reproduces $\binom{L}{\rho}$. This is an exact quotient-DAG accounting identity. It should not be

interpreted as a guarantee of compression: in the length-40 application every realized quotient multiplicity is 1.

Secondary finite comparison

The structural theorem does not predict which assigned blocks admit a completion. This appendix records one frozen finite comparison as an illustration of the target-loaded feasibility problem, not as evidence for the theorem itself.

For this section we use three explicit predicates.

- An A -block is **AF-compatible** if there exists a profile-correct F -block such that the complete $\{A, F\}$ subset gate on the cover word **faf** is passed.
- An (E, A) pair is **AFE-completable** if there exists a profile-correct F -block for which the coded factor **afe** contains no forbidden half-period $K \in [2, 40]$.
- An (E, A) pair is **jointly completable** if the **same** F -block satisfies both conditions.

The labels RX and H refer to two frozen deterministic populations. RX is the preregistered random-profile comparison population. H is the frozen canonical/historical comparison population assembled from earlier compatible constructions. Because the latter is selected by construction, the two populations are not random samples from a common distribution.

A preregistered deterministic quota $Q = 5000$ per E -word was applied in the persisted enumeration order. The clean RX run evaluated

75 111

(E, A) trials, representing 36 E -words. It produced 137 AF-compatible pairs from 17 productive E -words, with zero unresolved decisions. None of those 137 pairs was AFE-completable.

Applying the identical quota rule to the frozen H population gives:

population	trials	E represented	AF-compatible	AFE-completable	jointly completable
RX	75,111	36	137	0	0
H	31,775	9	263	86	44

The cleanest exact-equal-exposure strata are:

stratum	E	trials	AF-compatible	AFE-completable	joint
RX-5000-EQ	10	50,000	63	0	0
H-5000-EQ	4	20,000	78	36	24
RX-1000-EQ	21	21,000	45	0	0
H-1000-EQ	8	8,000	40	19	0

The AFE-completeness predicate was independently recomputed on all 263 quota-matched H pairs. Two implementations agreed on all 263 cases: 86 positive, 177 negative, and 0 unresolved. For every positive case, a separate direct checker validated one literal F -word against the substring Parikh condition; all 86/86 witnesses passed. Forty-two pairs are AFE-completable but not jointly completable, which is a direct control that the second implementation computes the AFE predicate rather than silently including the AF gate.

These counts are exhaustive only for the declared frozen finite populations. They are not probability estimates, and the zero count in RX is not a nonexistence theorem.

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